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H-JEPA: End-to-End Learning of Hierarchical World Models for Visual Planning

Stacking Action-Conditioned JEPAs Hierarchically Raises Visual AntMaze Success from 18% to 73% by Letting Each Level Predict Farther Ahead in Its Own Latent Space

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Flat world models trained at a single temporal resolution conflate the timescales of muscle-level control, motion planning, and goal-directed navigation, forcing planners to reason about irrelevant visual detail. H-JEPA stacks action-conditioned JEPAs so that higher levels automatically discard fast-varying sensory noise and retain the slow, task-relevant state that long-horizon planning depends on—without any manual specification of sub-goals or timescale boundaries.

AT A GLANCE

Problem: Single-level world models force planners to reason at one temporal resolution, wasting capacity on irrelevant low-level detail during long-horizon planning.

Why it matters: Robotics, navigation, and embodied AI require simultaneous reasoning across sub-second control and minute-scale goals.

Method: Stack L action-conditioned JEPAs; each level predicts k steps ahead in the latent of the level below; train all levels jointly end-to-end with a VICReg-style regularizer.

Result: A 3-level hierarchy raises Visual AntMaze success from **18% to 73%** while reducing planner compute.

Evidence: Visual AntMaze, robotic manipulation, and real-robot planning on the DROID dataset.

The Big Picture

Long-horizon planning requires simultaneous reasoning across radically different timescales: sub-second muscle control, second-scale motion planning, and minute-scale goal navigation. A flat world model trained at a single temporal resolution cannot separate these concerns—it must predict every future state at the same resolution, forcing the planner to process irrelevant low-level visual detail even when making high-level route decisions.

H-JEPA (Hierarchical Joint Embedding Predictive Architecture) resolves this by stacking multiple levels of action-conditioned JEPAs, each operating at a coarser timescale than the one below. The hierarchy is not hand-designed: timescale separation emerges automatically from end-to-end training when the world contains factors that evolve at different speeds.

Why Existing Approaches Fall Short

Flat world models are computationally wasteful.

A single-level model predicts every future state at native resolution, burdening the planner with fine-grained sensory detail even during high-level decision making.

Hand-designed hierarchies require manual sub-goals.

Options frameworks and HIRO-style methods pre-specify the timescale structure and require human-designed sub-goal spaces, sacrificing end-to-end trainability.

Generative hierarchical models are fragile.

Pixel-prediction hierarchies accumulate errors across levels and are sensitive to the choice of reconstruction loss.

Flat JEPA models remain at a single timescale.

V-JEPA and related work learn expressive single-level latent predictors; extending them to multiple timescales in an end-to-end trainable manner is non-trivial.

Core Mechanism

H-JEPA stacks L levels of action-conditioned JEPAs:

- **Level 1** operates at the native frame rate, predicting the latent of the next frame from the current latent and action.
- **Level $\ell > 1$** operates at a k -times-coarser timescale. It receives the temporally pooled latent from level $\ell-1$ and predicts that latent k frames ahead.
- **End-to-end training** jointly minimizes a JEPA objective at all levels: minimize distance between predicted latent and the stop-gradient target from the level below, regularized by a VICReg-style term to prevent representational collapse.

The key emergent property is *timescale separation*: higher levels learn to discard fast, unpredictable visual detail and retain slow, task-relevant state (object positions, goal proximity) without explicit supervision.

Technical Formulation

Let z_t^ℓ denote the latent at level ℓ and time t . Temporal pooling P_k averages k consecutive latents. The H-JEPA target at level ℓ is:

$$z_t^\ell = P_k \left(z_{t-k}^{\ell-1}, \dots, z_{(t+1) \cdot k - 1}^{\ell-1} \right)$$

The predictor f_ϕ^ℓ at level ℓ predicts k steps ahead:

$$\hat{z}_{t+1}^\ell = f_\phi^\ell(z_t^\ell, a_{t:k:(t+1) \cdot k})$$

The per-level JEPA loss is:

$$\mathcal{L}^\ell = \|\hat{z}_{t+1}^\ell - \text{sg}(z_{t+1}^\ell)\|^2 + \lambda \cdot \mathcal{L}_{\text{VICReg}}(z^\ell)$$

The total objective sums over all levels:

$$\mathcal{L}_{\text{H-JEPA}} = \sum_{\ell=1}^L \mathcal{L}^\ell$$

Planning applies a latent MPPI or MCTS planner at the highest level L , scoring rollouts in the level- L latent space—dramatically reducing compute per planning node relative to planning at level 1.

Learning or Inference Procedure

Training:

1. Collect trajectories $(o_1, a_1, \dots, o_T, a_T)$.
2. Encode observations to $z^0 = g_\psi(o)$; propagate up: $z^\ell = P_k(z^{\ell-1})$.
3. Train all levels jointly with $\mathcal{L}_{\text{H-JEPA}}$.
4. Optionally add inverse-dynamics heads at each level.

Planning at test time:

1. Encode current observation; propagate to level L .
2. Run MPPI or MCTS in the level- L latent space.
3. Execute the first action; receive next observation; repeat.

What the Guarantee Says

No formal convergence guarantee is provided. The central empirical claim is that timescale separation emerges automatically from end-to-end training when the data contains factors that evolve at different speeds. This is validated by probing each level’s representations for factors of variation at different timescales. Ablations confirm that a flat JEPA with the same total parameter count does not match the hierarchical planner.

Experimental Findings

- **Visual AntMaze:** A 3-level hierarchy raises success rate from **18% (flat JEPA) to 73%** while requiring less planner compute.
- **Manipulation:** H-JEPA outperforms flat JEPA on robotic pick-and-place and peg-insertion tasks requiring multi-step coordination.
- **DROID (real robot):** With inverse-dynamics supervision, H-JEPA improves offline planning fidelity on diverse real-robot demonstration data.

- **Timescale analysis:** Level-3 representations encode task-relevant state (goal proximity, object position) almost entirely; level-1 is dominated by low-level visual texture.

Ablations and Interpretation

Number of levels: Performance improves consistently from 1 to 3 levels; 4 levels adds marginal gain but increases training complexity.

Pooling stride k : Smaller strides ($k=2$) help tasks with fast-changing task-relevant state; larger strides ($k=8$) help tasks with very long planning horizons.

Timescale separation: Probing confirms the hierarchy: level 1 is dominated by visual texture; level 3 is nearly entirely task-state driven.

Comparison to hand-designed hierarchies: H-JEPA matches or exceeds options-based and HIRO-style methods on standard benchmarks while requiring no manual specification of sub-goals or timescale boundaries.

Reference

Wancong Zhang, Basile Terver, Michael Rabbat, Yann LeCun & Randall Balestriero. **H-JEPA: End-to-End Learning of Hierarchical World Models for Visual Planning.** arXiv:2610.06805, October 2026.
<https://arxiv.org/abs/2610.06805>